STATISTICS 10 · IQR / BOX PLOTS

CLASS 2

# FROM CONSTRUCTION TO INTERPRETATION

Fences · Outliers · Whiskers · Spread · Shape · Comparison

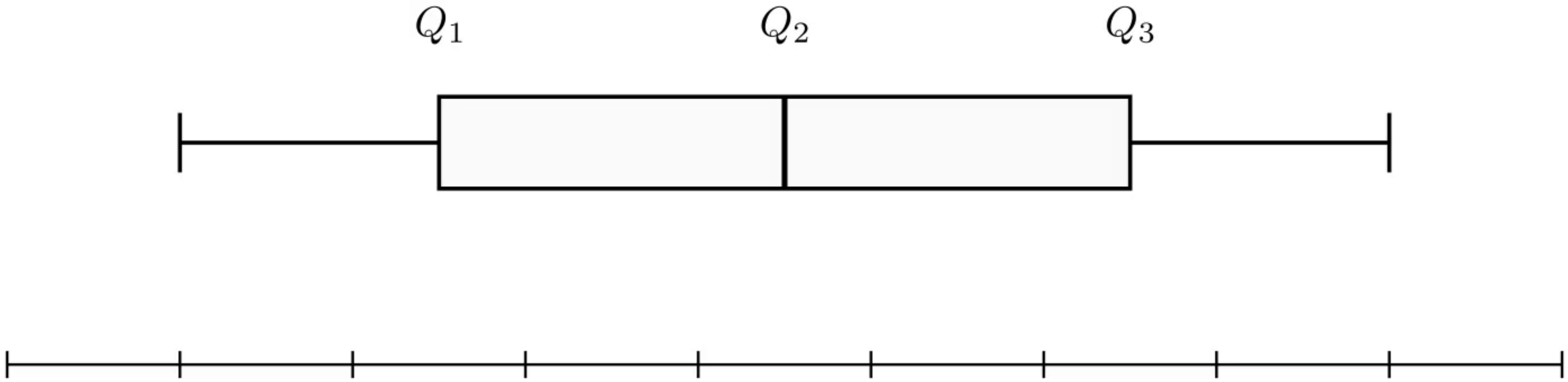
| | | | | | |
|---|---|---|---|---|---|
| 1 ORDER | 2 QUARTILES | 3 IQR | 4 BOXPLOT | 5 INTERPRET | 6 COMPARE |

Last class we learned how to build the graph. Today we learn how to read it.

# 01 RECALL: WHAT DO WE ALREADY KNOW?

Read the box before adding a new rule: Q2 describes center and Q3 - Q1 describes the middle 50%.

## But what happens when one value is very far away?

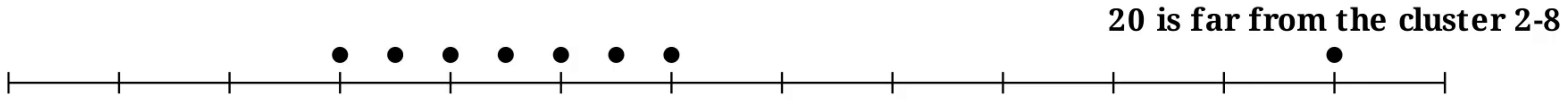



| | | |
|---|---|---|
| $Q_2 \rightarrow$ center | $Q_3 - Q_1 \rightarrow$ middle 50% | whiskers $\rightarrow$ extreme regular values |

# 02 WHY FENCES EXIST

Fences are decision limits used to classify unusually distant observations; they are not data values we invent.

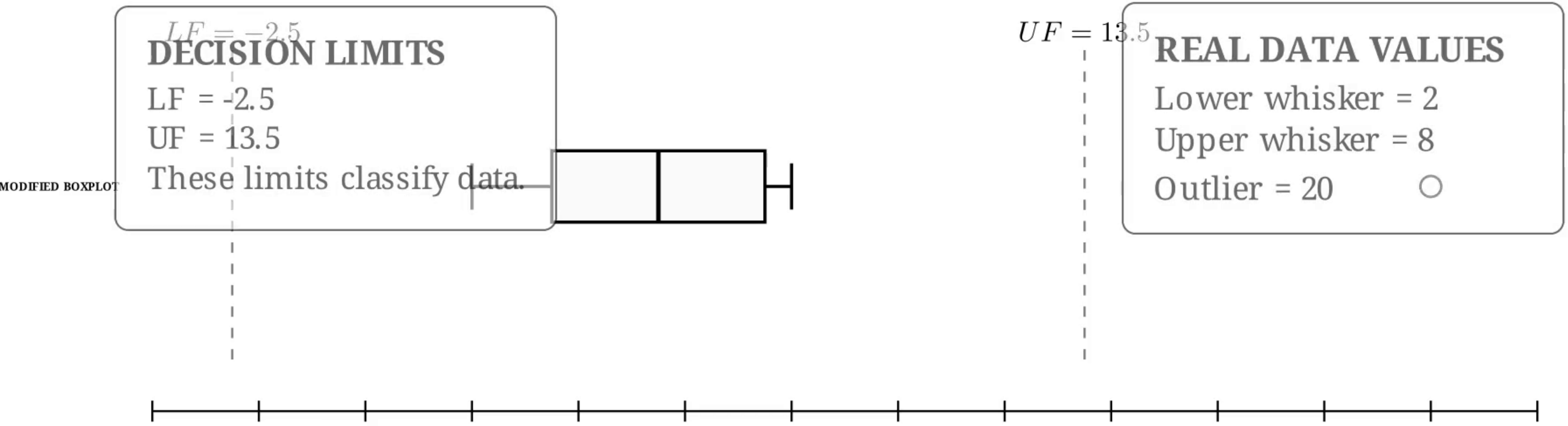
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 3 | 4 | 5 | 6 | 7 | 8 | 20 |




**Should the whisker automatically extend all the way to 20?**

# 03 FENCES ARE NOT WHISKER ENDPOINTS

A modified boxplot stops each whisker at a real non-outlier observation and plots observations beyond the fences separately.



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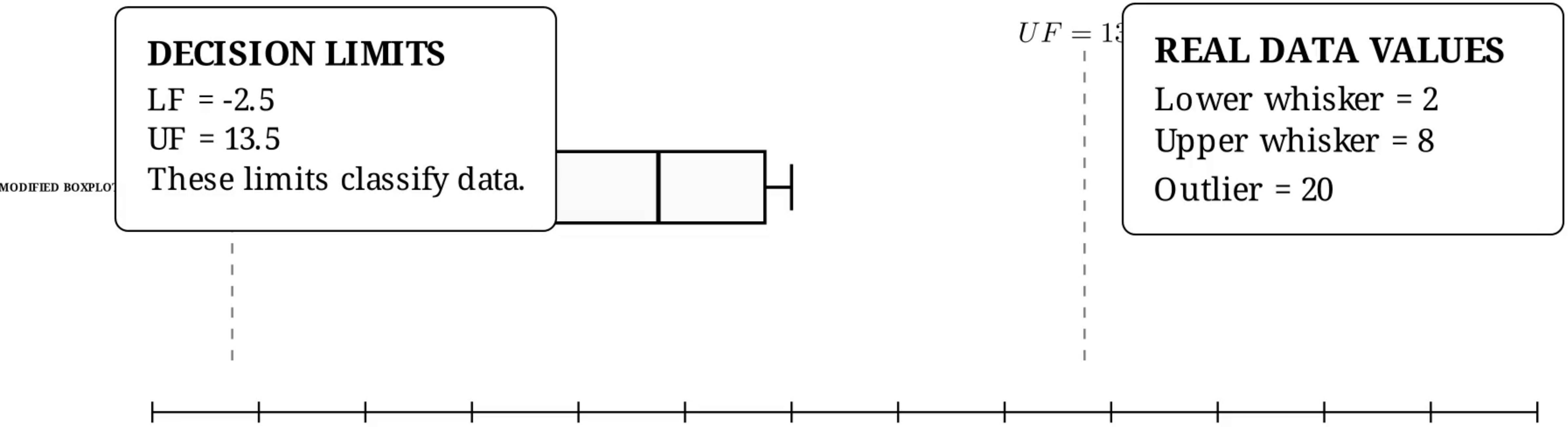



whisker = most extreme REAL observation still inside the fences

# 04 WORKED EXAMPLE: CLASSIFY BEFORE DRAWING

The arithmetic creates the decision rule; the ordered data then determines the actual whisker endpoints.

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 2 | 3 | 4 | 5 | 6 | 7 | 8 | 20 |

$$IQR = 7.5 - 3.5 = 4$$

$$LF = 3.5 - 1.5(4) = -2.5$$

$$UF = 7.5 + 1.5(4) = 13.5$$

$$20 > 13.5 \;\Rightarrow\; 20 \text{ is an upper outlier}$$

**NOW READ THE DATA**

Smallest regular value: 2

Largest regular value: 8

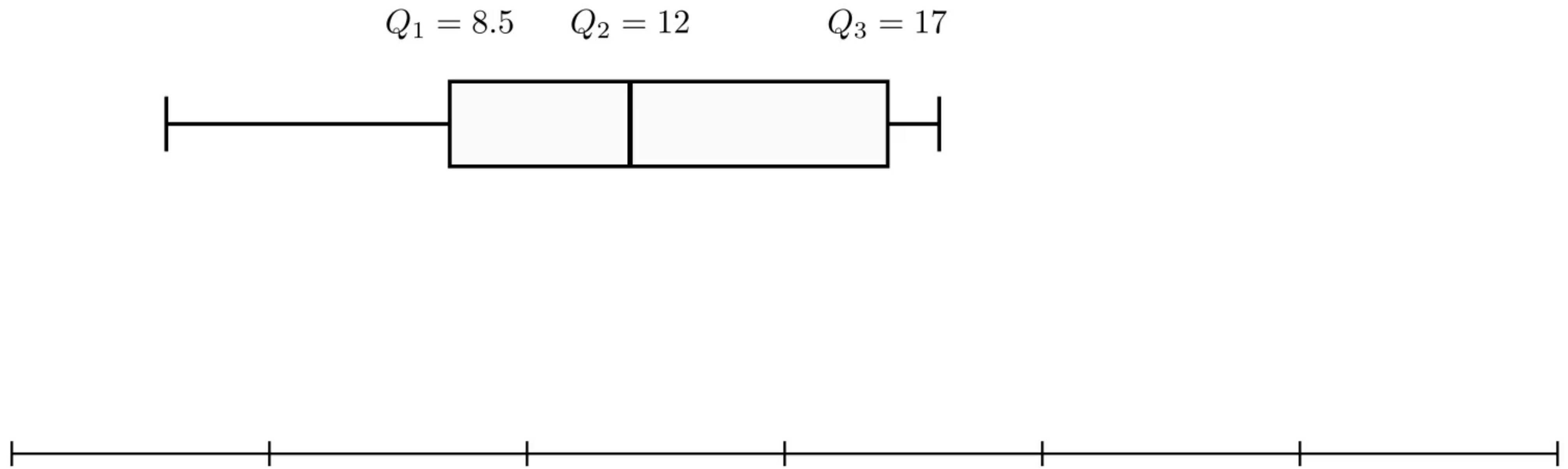
20 is drawn as a separate point.

The whisker does NOT end at 13.5.

# 05 READ THE GRAPH, NOT ONLY THE FORMULA

A boxplot communicates center, middle-50% spread, regular extremes and unusual observations on one numerical scale.

**EXAMPLE**




**CENTER**

Median = 12

**MIDDLE 50%**

IQR = 8.5 units

Half the data lie inside the box.

**REGULAR EXTENT**

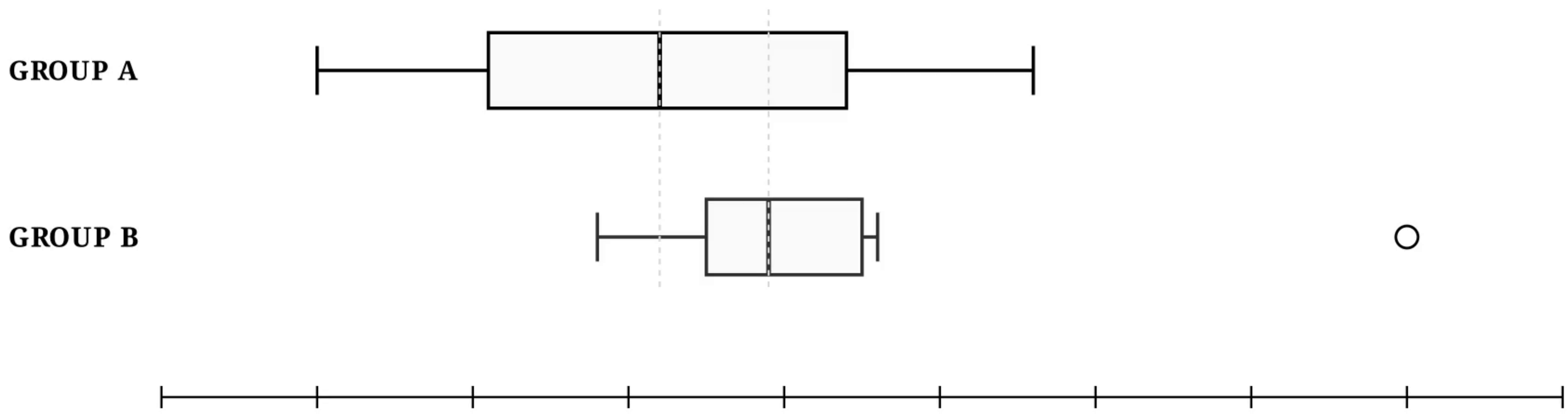
Whiskers: 3 to 18

Outlier: 35

# 07 COMPARE CENTER FIRST

Median position answers a center question; do not confuse a higher median with greater variability.

A: 40, 44, 47, 50, 52, 55, 59, 63
B: 49, 52, 53, 54, 55, 57, 58, 75



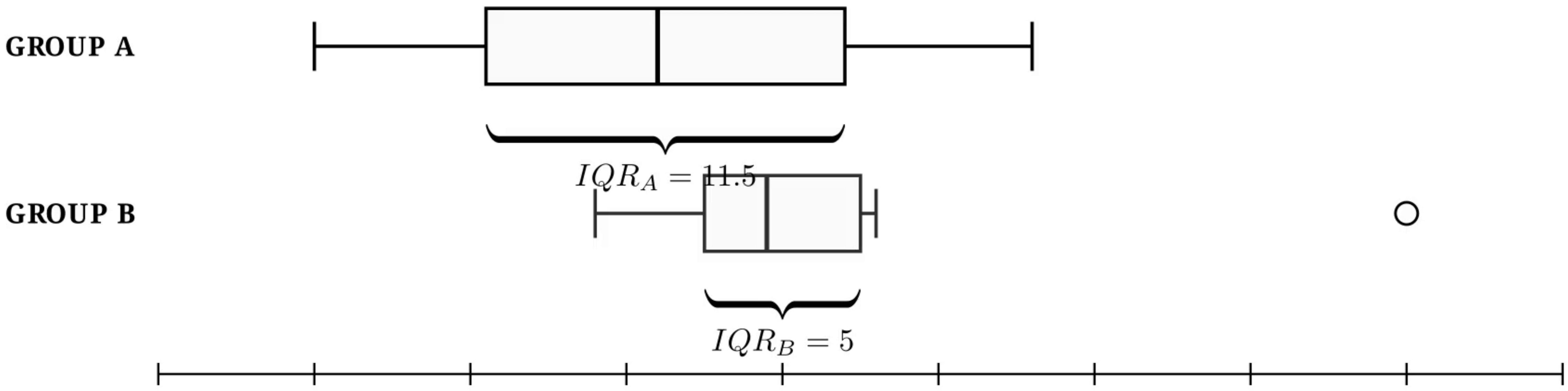

**SAME SCALE · 35 to 80**

$$Q_{2,B} = 54.5 > Q_{2,A} = 51 \quad \Rightarrow \quad \text{Group B has the higher median.}$$

# 08 COMPARE THE MIDDLE 50% WITH IQR

The box width is meaningful only because both boxplots use the same numerical scale.

A: 40, 44, 47, 50, 52, 55, 59, 63
B: 49, 52, 53, 54, 55, 57, 58, 75



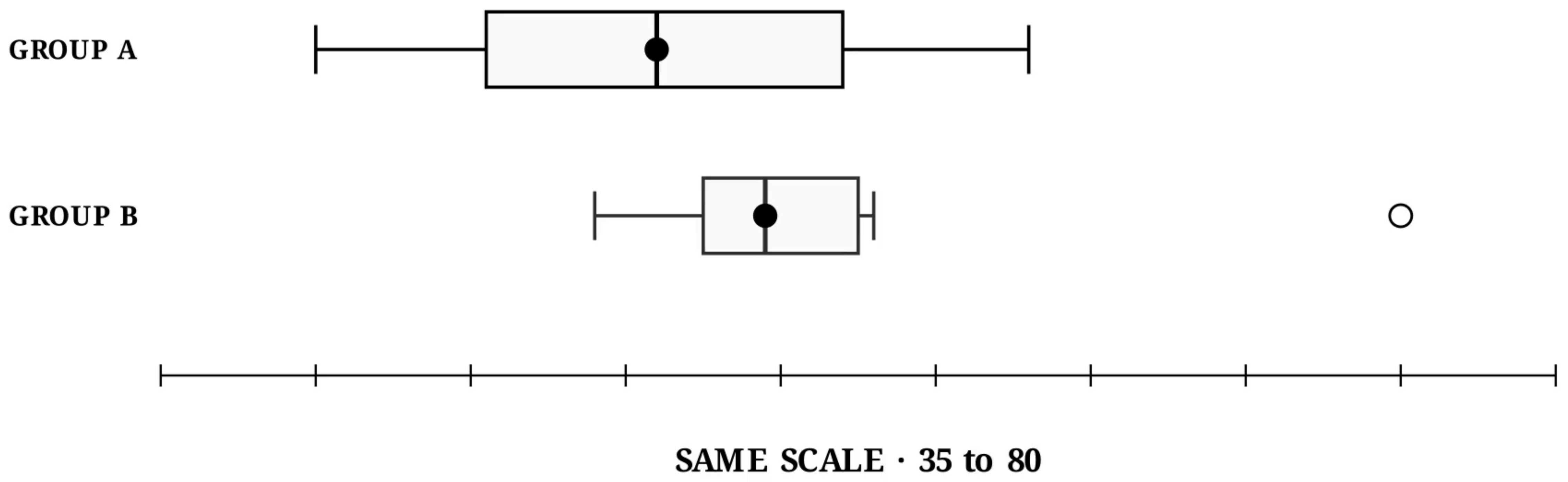

**SAME SCALE · 35 to 80**

**Group B has a smaller IQR, so its central observations are more tightly clustered.**

# SHAPE CLUES: BE CAUTIOUS

Unequal box halves or whiskers can suggest asymmetry, but a boxplot is a compact summary - not a complete picture of distribution shape.

**Group A looks roughly balanced around its median.**
**Group B shows an upper-side irregularity - a clue, not proof of shape.**



# 12 TURN NUMBERS INTO A STATISTICAL CONCLUSION

A strong comparison names the feature, cites evidence and explains what the evidence means in context.

**CENTER**

Group B has the higher median: 54.5 compared with 51.

**MIDDLE 50%**

Group B spans only 5 units. Group A spans 11.5 units.

**QUALIFICATION**

Group B also contains one unusually high value: 75.

**Therefore: Group B is more tightly clustered in its central 50%, but it also has an unusually high observation.**

# 13 COMMON MISTAKES TO AVOID

Each mistake changes the statistical meaning of the graph or makes a comparison invalid.

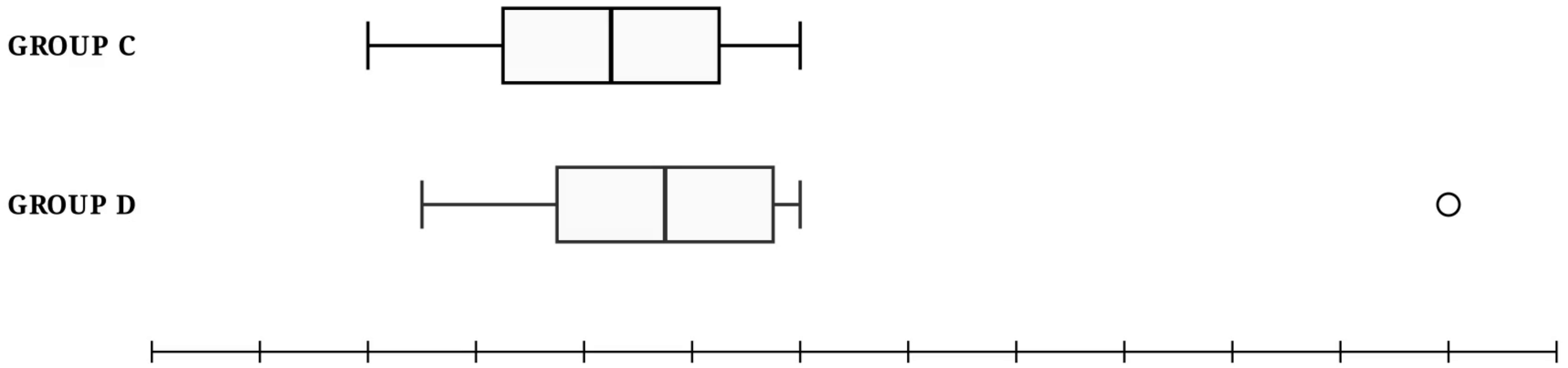
1. **USING A FENCE AS A WHISKER**
2. **CONNECTING AN OUTLIER TO THE WHISKER**
3. **COMPARING PLOTS WITH DIFFERENT SCALES**
4. **SAYING ONLY: 'THIS GRAPH IS BIGGER'**

Compare: center + IQR + regular spread + asymmetry clues + outliers

# 14 YOUR TURN: COMPARE BEFORE THE REVEAL

Use the same order every time: center -> IQR -> whiskers -> possible asymmetry -> outliers -> conclusion.

**1) Which group has the higher median? 2) Compare IQR. 3) Identify any outlier.**




$Q_{2,D} = 17.5 > Q_{2,C} = 16.5$

$IQR_C = IQR_D = 4$

$32 > UF_D = 25.5 \Rightarrow$ upper outlier

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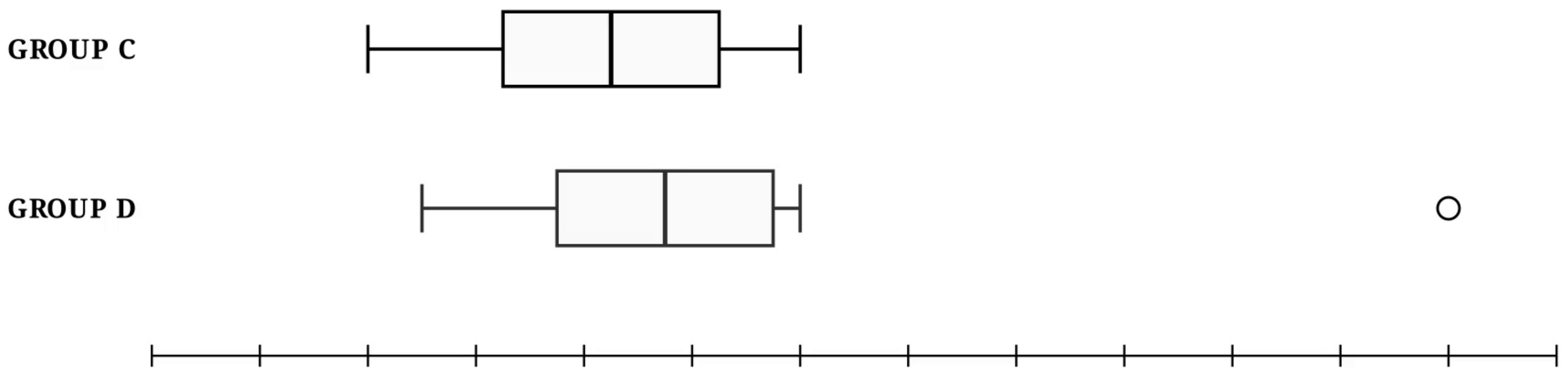


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# 16 NEXT: FROM QUARTILES TO MORE POSITION MEASURES

Today ends with interpretation. The next lesson generalizes position using deciles and percentiles.

| 1 QUARTILES | 2 DECILES | 3 PERCENTILES |
|---|---|---|
| $Q_1 = P_{25}$ | $Q_2 = P_{50}$ | $Q_3 = P_{75}$ |

**Class 2 takeaway: construct correctly, then compare with evidence and precise statistical language.**

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